GROUP 6:

COMPUTER NETWORKS AND INTERNET


POV
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WHAT IS A COMPUTER NETWORK?

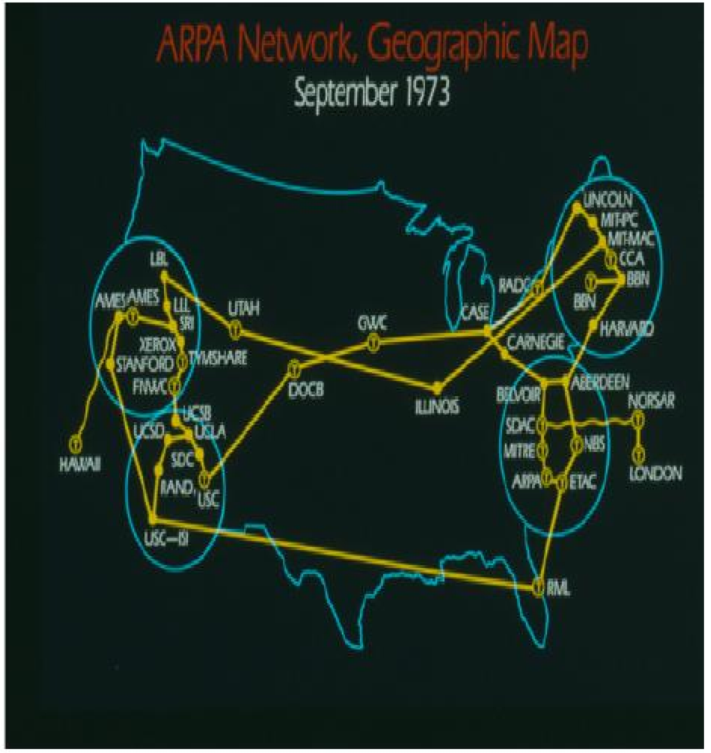
- In information technology, a network is defined as **the connection of at least two computer systems**, either by a cable or a wireless connection.
- The main task of a network is to provide participants with a single **platform** for **exchanging data** and sharing **resources**. This task is so important that many aspects of everyday life and the modern world would be unimaginable without networks

The main advantages of networks are:

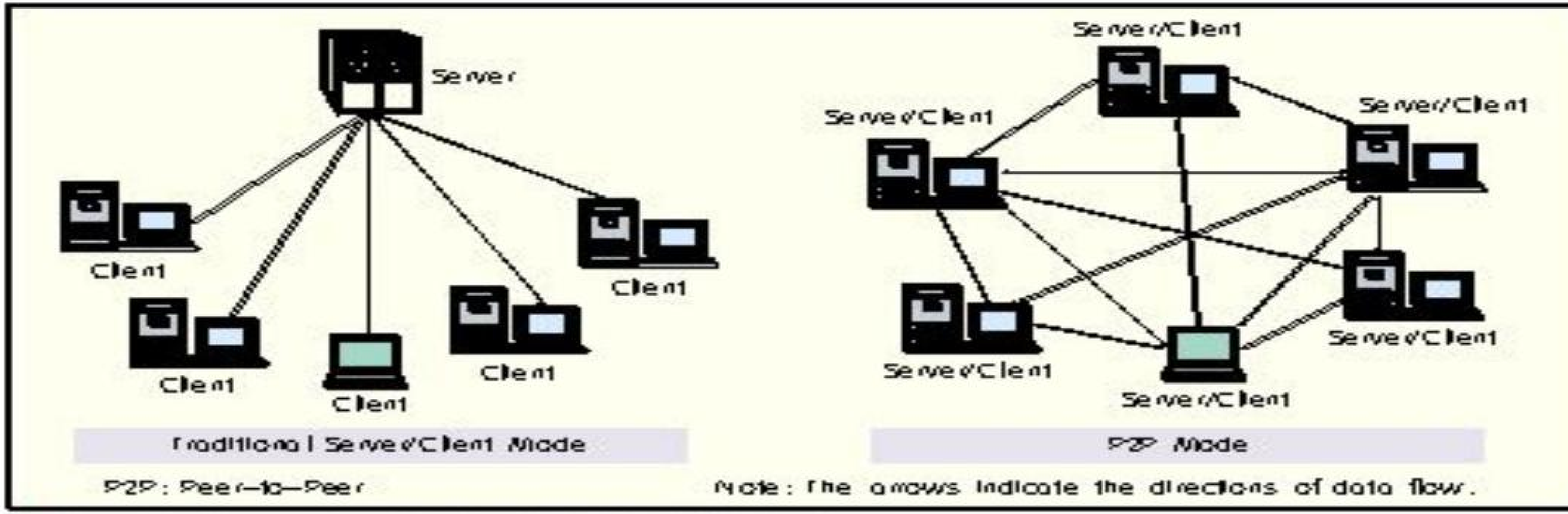
- Shared use of data
- Shared use of resources
- Central control of programs and data
- Central storage and backup of data
- Shared processing power and storage capacity
- Easy management of authorizations and responsibilities

TRIVIA: ARPANET

- The U.S **Advanced Research Projects Agency Network (ARPANET)** was the first public packet-switched computer network. It was first used in 1969 and finally decommissioned in 1989. ARPANET's main use was for academic and research purposes.

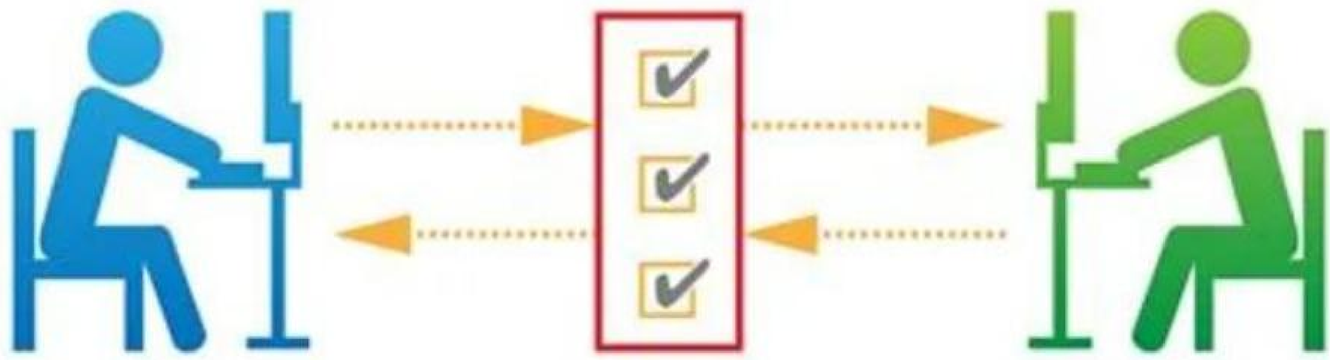


HOW DOES A NETWORK WORK?

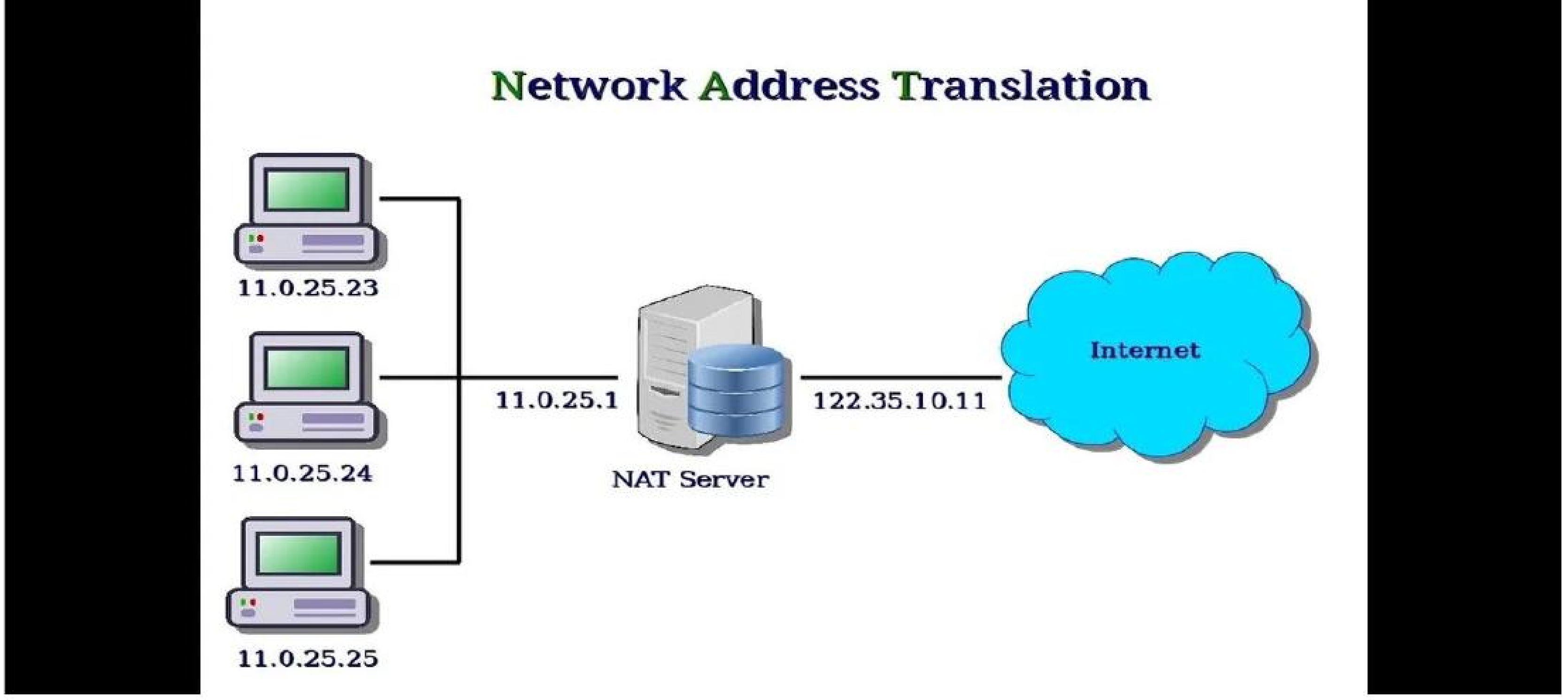


▲Figure 1. Comparison between P2P and C/S modes.

- In a typical client-server network there is a central node called the **server**. The server is connected to the other devices, which are called **clients**. This connection is either wireless (**Wireless LAN**) or wired (**LAN**). The server and the client communicate as follows in this server-based network: The client first sends a **request** to the server. The server evaluates the request and then transmits the **response**. In this model, the client always connects to the server, never the other way around.



Network Protocols- Network protocols ensure smooth communication between the different components in a network. They control data exchange and determine how communication is established and terminated as well as which data is transmitted.



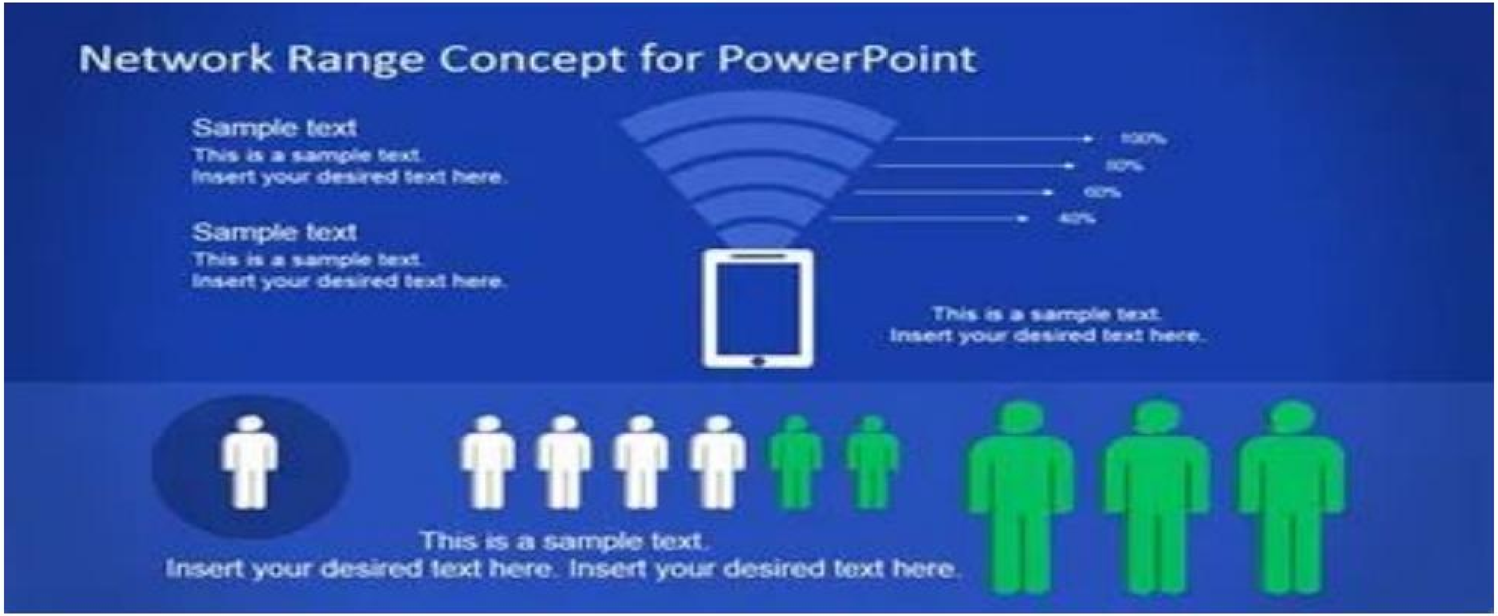
- **Network Addresses-** Network addresses are used to ensure that the transmitter and receiver can be correctly identified. This is also known as the **Internet Protocol address (IP address).**

Internet Service Provider (ISP)- An Internet service provider is an organization that provides services for accessing, using, or participating in the Internet. ISPs can be organized in various forms, such as commercial, community-owned, non-profit, or otherwise privately owned.

TYPES OF NETWORKS

- **Wireless vs. wired**

Networks are classified by transmission type as either wireless or wired. Examples of wireless networks include Wi-Fi networks based on the IEEE 802.11 standard, or the LTE networks used for mobile devices and smartphones. Wired networks such as DSL are also known as broadband Internet.

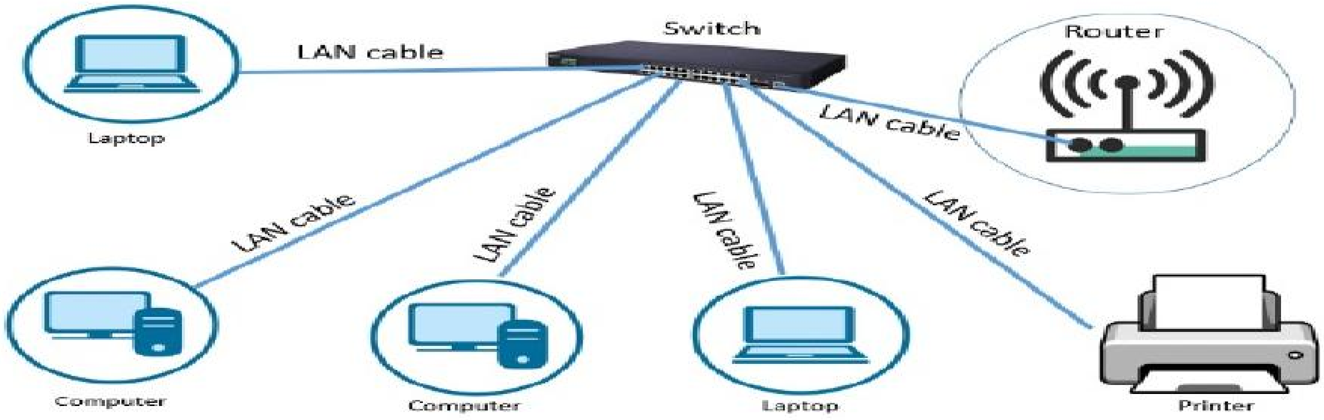


- **<u>Network range</u>**
- Networks are typically classified by range as follows

Personal Area Network (PAN)

- Is used for interconnecting devices within a short range of **approximately 10 meters**
- Examples include Bluetooth technology or Apple’s Airdrop ad hoc Wi-Fi service

Local Area Network (LAN)



Local Area Network

- Refers to networks **within a limited area**
- Are among the most widespread networks and are used in households or small and medium-sized companies
- is limited between **100-1000 meters** coverage

RJ 45 AND LAN CABLE

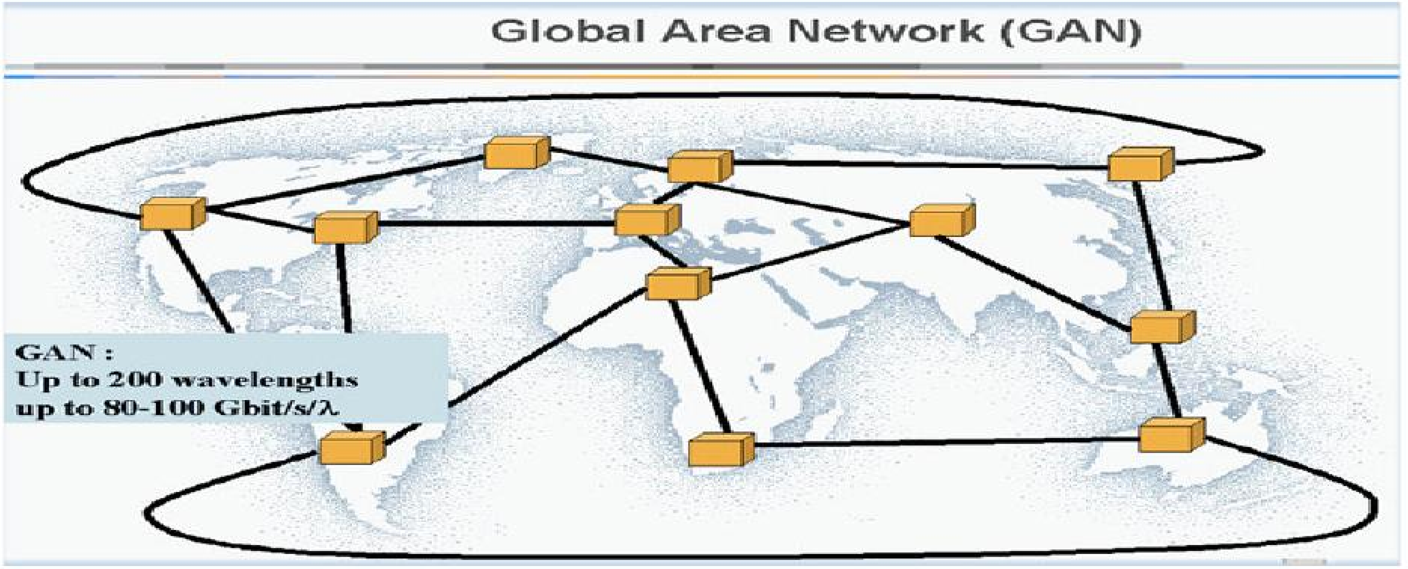
Metropolitan Area Network (MAN)

- A network that spans a large area, such as a town, cities, or single geographic regions
- An example of a MAN is a series of wireless routers distributed across a city
- Usually ranges from **5 kilometers to 50 km**

Wide Area Network (WAN)

- Are **<u>large-scale networks</u>** that can stretch over countries and even continents
- These networks cover large geographical regions and link different smaller networks like LAN's or MAN's with one another
- Can range up to 100,000 km and in some cases, stretches globally or over international borders

Global Area Network (GAN)

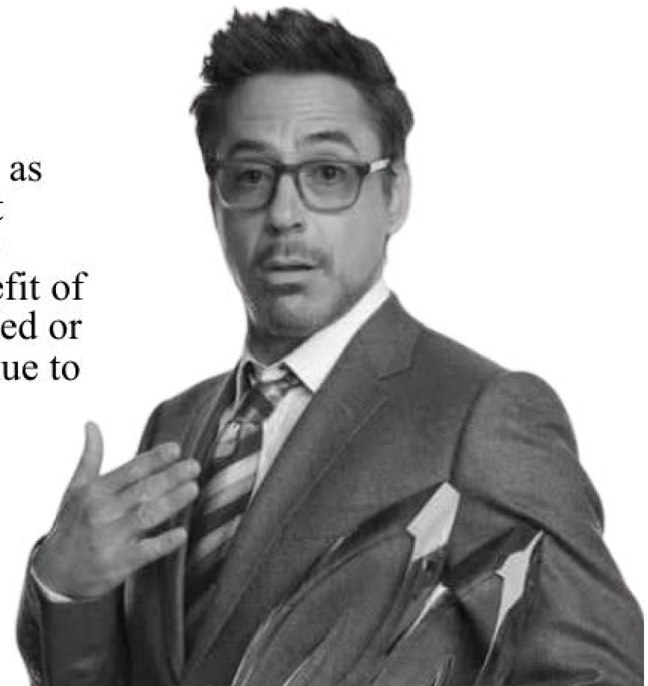


- Refers to a network composed of different interconnected networks that cover an unlimited geographical area
- The term is loosely synonymous with Internet, which is considered a global area network

WHAT IS THE INTERNET?

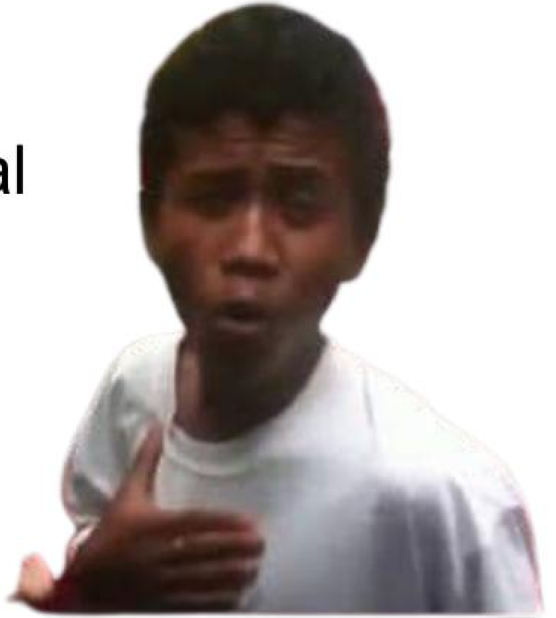
The Internet, sometimes called simply "**the Net**," is a worldwide system of computer networks -- a network of networks in which users at any one computer can, if they have permission, get information from any other computer (and sometimes talk directly to users at other computers).

It was conceived by the Advanced Research Projects Agency (ARPA) of the U.S. government in 1969 and was first known as the ARPANET. The original aim was to create a network that would allow users of a research computer at one university to "talk to" research computers at other universities. A side benefit of **ARPANet's** design was that, because messages could be routed or rerouted in more than one direction, the network could continue to function even if parts of it were destroyed in the event of a military attack or other disaster.



ADVANTAGES OF THE INTERNET

- Today, the Internet is a public, cooperative and self-sustaining facility accessible to hundreds of millions of people worldwide. It is used by many as the primary source of information consumption, and fueled the creation and growth of its own social ecosystem through social media and content sharing. Furthermore, e-commerce, or online shopping, has become one of the largest uses of the Internet.



The main advantages of the internet are:

- Information, knowledge, and learning
- Connectivity, communication, and sharing
- Address, mapping, and contact information
- Banking, bills, and shopping
- Selling and making money
- Collaboration, work from home, and access to a global workforce
- Donations and funding
- Entertainment
- Internet of things
- Cloud computing and cloud storage

HOW DOES THE INTERNET WORK?

- Physically, the Internet uses a portion of the total resources of the currently existing public telecommunication networks. Technically, what distinguishes the Internet is its use of a set of protocols called Transmission Control Protocol/Internet Protocol (TCP/IP). Two recent adaptations of Internet technology, the Intranet and the extranet, also make use of the TCP/IP protocol.
- The Internet can be seen as having two major components: **network protocols** and **hardware**. The **protocols**, such as the TCP/IP suite, present sets of rules that devices must follow in order to complete tasks. Without this common collection of rules, machines would not be able to communicate.The protocols are also responsible for translating the alphabetic text of a message into electronic signals that can be transmitted over the Internet, and then back again into legible, alphabetic text.

- **Hardware**, the second major component of the Internet, includes everything from the computer or smartphone that is used to access the Internet to the cables that carry information from one device to another. Additional types of hardware include satellites, radios, cell phone towers, routers and servers. These various types of hardware are the connections within the network. Devices such as computers, smartphones and laptops are **end points, or clients**, while the machines that store the information are the **servers**. The transmission lines that exchange the data can either be wireless signals from satellites or 4G and cell phone towers, or physical lines, such as cables and fiber optics.

- The process of transferring information from one device to another relies on packet switching. Each computer connected to the Internet is assigned a unique IP address that allows the device to be recognized. When one device attempts to send a message to another device, the data is sent over the Internet in the form of manageable packets. Each packet is assigned a port number that will connect it to its endpoint.

- A packet that has both a unique IP address and port number can be translated from alphabetic text into electronic signals by travelling through the layers of the OSI model from the top application layer to the bottom physical layer. The message will then be sent over the Internet where it is received by the Interne service provider's Internet Service Provider (ISP) router. The router will examine the destination address assigned to each packet and determine where to send it.

- Eventually, the packet reaches the client and travels in reverse from the bottom physical layer of the OSI model to the top application layer. During this process, the routing data -- the port number and IP address -- is stripped from the packet, thus allowing the data to be translated back into alphabetic text and completing the transmission process.

WORLD WIDE WEB

- The Web is the most widely used part of the Internet. Its outstanding feature is hypertext, a method of instant cross-referencing. In most Web sites, certain words or phrases appear in text of a different color than the rest; often this text is also underlined. When a user selects one of these words or phrases, they will be transferred to the related site or page. Buttons, images, or portions of images are also used as hyperlinks.
- The Web provides access to billions of pages of information. Web browsing is done through a Web browser.

WEB BROWSER

- A web browser (also referred to as an Internet browser or simply a browser) is application softaware for accessing the World Wide Web or a local website

POPULAR WEB BROWSERS

- **GOOGLE CHROME**

-**A cross-platform web browser developed by Google was first released in 2008 for Microsoft Windows, built with free software components from Apple WebKit and Mozilla Firefox. It was later ported to Linux, macOS, iOS, and Android, where it is the default browser**

Mozilla Firefox

- Created by Dave Hyatt and Blake Ross is a free and open-source web browser developed by the Mozilla Foundation and its subsidiary, the Mozilla Corporation
- first released as Firefox 1.0 on November 09, 2004

Internet Explorer

- is a discontinued series of graphical web browsers developed by Microsoft which was used in the Windows line of operating systems
- first released in 1995 as part of the add-on package Plus! for Windows 95 that year
- replaced by Microsoft Edge.

UNIFORM RESOURCE LOCATOR (URL)

- A **URL (Uniform Resource Locator**) is a unique identifier used to locate a resource on the Internet. It is also referred to as a web address. URLs consist of multiple parts -- including a protocol and domain name -- that tell a web browser how and where to retrieve a resource.

TRIVIA: INFO.CERN.CH

- The first ever website established by Tim Berners-Lee

THANK YOU FOR LISTENING !!!!

- MORAL LESSON?

I MISS YOU haha

GAME CONSOLES

GROUP 7

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Aristotle Sandulan

Dennis John Bisaro

WHAT IS GAME CONSOLE?

A video game console is a computer device that outputs a video signal or visual image to display a video game that one or more people can play. ... The term "video game console" is primarily used to distinguish a console machine primarily designed for consumers to use for playing video games, in contrast to arcade machines or home computers.

Different type of Game Console

- **Home GAME Console**
- **Handheld** GAME **Console**
- **Hybrid** GAME **Console**

Home Game Console - Home Game Console is a video game console that is designed to be connected to a display device, such as a television, and an external power source as to play video games. Home Game Consoles are generally less powerful and customizable than personal computers, designed to have advanced graphics abilities but limited memory and storage space to keep the units affordable.

Handheld Game Console - Handheld Game Console is a small, portable self-contained video game console with a built-in screen, game controls and speakers. Handheld game consoles are smaller than home video game consoles and contain the console, screen, speakers, and controls in one unit, allowing people to carry them and play them at any time or place.

Hybrid Game Console - A Hybrid Console is a gaming platform that can be used in more than one way. For example, the Nintendo Switch can be used as a handheld console but can also be plugged into a TV and played like an Xbox or PlayStation.

THE FATHER OF GAME CONSOLE

Ralph H. Baer

also known as THE FATHER OF VIDEO GAMES

The first Game Console invented by Ralph Baer

The Magnavox Odyssey (1972)

The Magnavox Odyssey is the first and oldest home video game console in the world. This home video game system was based on the "Brown Box" prototype invented by Ralph Baer, who is considered the Father of the Video Game.

The first Handheld Game Console

MICROVISION(1979)

The **Microvision** (aka **Milton Bradley Microvision** or **MB Microvision**) is the first handheld game console that used interchangeable cartridges and in that sense is reprogrammable. It was released by the Milton Bradley Company in November (1979

(1977) Atari 2600

- Atari changed console gaming forever in 1977 by creating the first popular console capable of accepting game cartridges. Before this, consoles (including Atari's own Pong) only had self-contained games. The VCS 2600 let players use joysticks or paddle controllers to play several hit games like Pacman, Asteroids, and Frogger.
- Developed by Atari, Inc.
- Atari, Inc. was an American video game developer and home computer company founded by Nolan Bushnell and Ted Dabney.

(1979) Intellivision

- In 1979, Mattel Electronics released their own console as a direct competitor to the Atari 2600. The Intellivision (or intelligent television), never surpassed the Atari in popularity, but it did offer competitive audio effects and graphics. Popular games on the Intellivision included golf games, and Las Vegas Poker & Blackjack.
- Manufacturer: Mattel Electronics(1979-1984) and INTV Corporation(1984-1990)

(1982) ColecoVision

- Coleco Industries released their competitor to the **Atari** and **Intellivision** consoles in 1982. The ColecoVision offered the ability to play games like Nintendo's Donkey Kong, Sega's Zaxxon, and some less-popular games like Lady Bug and Cosmic Avenger. The Coleco also offered expansion modules that allowed gamers to play popular games from Atari, racing games, and more.
- ColecoVision was discontinued in 1985 when Coleco withdrew from video game market.

(1985) Nintendo Entertainment System

•The Nintendo Entertainment System (NES) was first introduced in Japan as the “Famicom,” but the console was rebranded as the NES in 1985 and introduced to the U.S. gaming market. Super Mario Bros launched the console into popularity. The NES was also the birthplace for many other legendary game franchises, including Nintendo’s Zelda, Metroid, and Square’s Final Fantasy

(1995) PlayStation

- In 1995, PlayStation commonly known as PS One
- The CD ROM capabilities of the console allowed games to have 3D graphics and more complex games, leading to popular a surge in new games like Gran Turismo, Metal Gear Solid, and Resident Evil.
- The console was primarily designed by Ken Kutaragi and Sony Computer Entertainment in Japan.

(1996) Nintendo 64

- The Nintendo 64 was one of the first game consoles to use a 64-bit processor, which gave gamers better graphics than ever before from a Nintendo system. Some hit games on the N64 included Goldeneye 007, Donkey Kong 64, Star Fox 64, and Super Mario 64.
- It was the last major home console to use cartridges as its primary storage format until the Nintendo Switch.

(2000) PlayStation2

- In 2000, Sony upped their game yet again with the PlayStation 2. The new console featured DVD playback abilities, which was a huge selling point at the time. It also offered backward compatibility, so gamers could still play their PS1 games. Some of the top games for this console included the Grand Theft Auto franchise, Shadow of the Colossus, and Kingdom Hearts.
- The PS2 offered backward-compatibility for its predecessor's DUALSHOCK controller, as well as its

Games.

(2001) Xbox

- In 2001, Microsoft hopping into the console realm and kicked off what's known as the "console wars." To compete with PlayStation, they introduced the Xbox Live service that let players compete online.

- The brand consists of five video game consoles, as well as applications, streaming services, an online service by the name of Xbox network, and the development arm by the name of Xbox Game Studios.

(2004)PlayStation Portable

The **PlayStation Portable**[a] (**PSP**) is a handheld game console developed and marketed by Sony Computer Entertainment.

The system was the most powerful portab

(2005) Xbox 360

- In continuation of the console wars, Microsoft threw another jab at Sony with the Xbox 360 in 2005. It featured improved graphics, wireless controllers, and third-party streaming capabilities.
- It competed with Sony's PlayStation and Nintendo Wii as part of seventh generation of video games consoles.

(2006) Nintendo Wii

- In the heat of the console war between Sony and Xbox, Nintendo released a family-friendly alternative. The Wii offered the ability to use motion to play games rather than simple buttons and joysticks. Players could bowl, play golf, and even duel with light sabers using a first-of-its-kind remote that helped get kids off the couch with active gameplay

(2006) PlayStation 3

•Sony released the PS3 as a competitor to the Xbox 360 in 2006. To compete with Xbox live, they introduced the PlayStation Network as a free alternative to Xbox live. Some popular games on the console included Assassin's Creed Rogue, Far Cry 4, Red Dead Redemption and more.

(2013) Xbox One

•Microsoft Xbox released their current generation of consoles in November 22, 2013 — the Xbox One. Like the Xbox 360, it continued to offer Xbox Live and third-party streaming. The new console boasted improved graphics, a sleek design, and support for a huge number of games. Some popular early Xbox One games included Call of Duty Black Ops III, Minecraft, and Fallout 4.

(2013) PlayStation 4

•Sony's current generation of PlayStation, the PS4, is the most advanced yet — introduced on the tail end of the Xbox One release. It has sharp graphics, VR capabilities, and an impressive library of games to choose from. The PS4 competes with the Nintendo Switch and Xbox One for popularity in 2018. Popular games for the console are Fortnite, Black Ops IV, and Destiny 2.

• Sony Computer Entertainment launch PS4 in November 2013

(2016)Xbox One S

- In 2016, Microsoft gave the Xbox One some major upgrades with the new Xbox One S. The updated console was smaller, could upscale games from 1080p(Progressive Scan) to 4K, stream 4K Ultra HD video and HDR(High Dynamic Range) Gaming support

(2016)PlayStation 4 Pro

•In 2016, Sony upgraded the PS4 to compete with 4K upscaling offered by the Xbox One S. The new PlayStation 4 Pro was smaller, sleeker, and could also handle 4K upscaling.

The PS4 PRO is the more powerful version of the PS4 console. Faster loading times, enhanced graphics, and higher framerates.

(2017) Xbox One X

- The current generation of Xbox was released in 2017, and offers even more graphics upgrades over the Xbox One S and PlayStation 4 Pro. The new Xbox One X supports 4k video and gameplay natively. It also boasts 40% more processing power than any other console on the market. Some current popular games for the Xbox One X include Fallout 76, NBA 2K19, Call of Duty: Black Ops 4, and Assassin's Creed Odyssey.

(2017)Nintendo Switch

•Nintendo changed the game in 2017 by offering a console that could do more. The new Nintendo Switch works as both a handheld console and a TV-based console, and the clip-in controllers give gamers a huge amount of flexibility.

• The console itself is a tablet that can either be docked for use as a home console or used as a portable device,making it a hybrid console.

(2020) PlayStation 5

•Released in November 2020, a full four years after its predecessor, PlayStation 5 helped introduce the ninth generation of video game consoles with two models: the PS5 with 4K Blu-ray drive for gamers who enjoy digital downloads, retail game support, and playback to their TVs and the Digital Edition model for gamers who rely solely on digital downloads. The console's Tempest 3D AudioTech affords it the capability to play hundreds of sound sources simultaneously while PS4 played only 50. Its DualSense wireless controllers feature immersive haptic feedback, adaptive triggers, and a built-in microphone. Higher refresh rates make game play feel smoother and a custom SSD gives PlayStation 5 the potential for the fastest game load times of any console on the market.

(2020) Xbox Series X and Series S

- The **Xbox Series X** and the **Xbox Series S** are home video game consoles developed by Microsoft. They were both released on November 10, 2020, as the fourth generation of the Xbox console family, succeeding the Xbox One. Along with Sony's PlayStation 5, also released in November 2020,

THE END